

Pavithra Kamatchi Soundaram

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EDUCATION

CORNELL UNIVERSITY

Ithaca, NY

Bachelor of Science, Computer Science (College of Engineering)

August 2023 - May 2027

Relevant Coursework: Data Structures & Algorithms, Computer Systems, Object-Oriented Programming, Machine Learning, AI

Reasoning, Linear Algebra, Statistics, Operating Systems, Natural Language Processing, Computer Networks

WORK EXPERIENCE

CORNELL UNIVERSITY (Hybrid Body Lab)

Ithaca, NY

AI Engineering Research Assistant

Feb 2026 – Present

- Engineered real-time multimodal pipelines for wearable and soft-robotic systems, synchronizing audio, IMU, and physiological sensor streams into stable feature representations.
- Built a few-shot multi-agent inference framework with parallel specialized LLM agents and a coordinator that integrates outputs using confidence-weighted consensus and robust feature-extraction/error-handling modules.
- Designed a deterministic state-mapping layer that transforms multimodal predictions into interpretable system modes for downstream interactive interfaces.

ARTICUL8 AI (Applied AI Research Team)

Santa Clara, CA

Software Engineering Intern

May 2024 – Aug 2024

- Developed modular Python code leveraging AWS S3 to automate extraction of images, captions, and text from scientific documents with diverse layouts and complex figures, enabling efficient dataset preparation for LLM training
- Engineered computer vision solutions using OpenCV and PyTesseract, achieving higher extraction accuracy and flexibility than Unstructured API across a diverse internal evaluation set

UNIVERSITY OF CHICAGO DATA SCIENCE INSTITUTE ([UCARE LAB](#))

Chicago, IL

Data Science Research Assistant

June 2022 – Aug 2022

- Predicted execution times for genomic files across 107 bioinformatic pipelines to reduce scheduling inefficiency and compute costs.
- Built XGBoost/Random Forest models to predict per-file execution times using genomic file metadata + compute/network features for cost-efficient cloud vs. on-prem routing.
- Improved scheduling accuracy by 19.5% through tuning and evaluation across 562K+ workflow records.
- Increased model reliability for heterogeneous workloads with standardized preprocessing and reproducible ML pipelines.

PROJECTS

CORNELL UNIVERSITY AUTONOMOUS DRONE PROJECT TEAM ([Link](#))

Oct 2023 - Present

Software Engineer (Computer Science Subteam)

- Built a custom ROS API for Gazebo simulation to automate drone takeoff, landing, and directional movement; collaborated with mechanical and ECE subteams to align simulation behavior with intended drone design.
- Implemented D\ Lite-based incremental path planning* for dynamic obstacle environments, enabling real-time replanning and stable navigation behavior in simulation.

FinSight - AGENTIC PERSONAL FINANCE ASSISTANT

July 2025 - Present

- Built a full-stack platform enabling transaction ingestion, natural-language querying, real-time analytics & anomaly detection.
- Engineered a multi-agent backend (NL-to-SQL, analytics, classification) with orchestrated LLM tool-calling, reducing latency 7× (96s → 11s) through caching and parallel execution.
- Developed a schema-aware NL-to-SQL engine with validation layers, SQL-operation restrictions, fallback logic, and detailed logging to ensure correctness and prevent hallucinations/injection.
- Designed secure multi-tenant infrastructure using Supabase RLS + JWT auth to guarantee isolated per-user data access.
- Built a responsive React frontend with CSV ingestion, dynamic charts, anomaly alerts, and intuitive budget-tracking workflows.

LEADERSHIP & OUTREACH

Chapter Co-President, Asha for Education – Raised over \$7,000 for a nonprofit student chapter supporting education for underprivileged children in India; organizing fundraisers and coordinating with partner schools to fund supplies and teacher salaries.

SKILLS

Technical Skills: Python, Java, MySQL, PostgreSQL, OCaml, HTML, CSS; LangChain, LLMs, AI Agents, Prompt Engineering, OpenAI API, FastAPI, REST API, SQLAlchemy, Supabase, TensorFlow, PyTorch, Scikit-learn, XGBoost, NumPy, Pandas; Matplotlib, Seaborn, OpenCV, Pytesseract, Librosa, Vosk, Pydub, Parselmouth, pdfplumber, LatexOCR; AWS S3; LaTeX, Streamlit, Git, ROS